BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Insertion Sort

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. When does ascending bubble sort swap two adjacent items?

2. How does the OS support multitasking?

A) Shares CPU time
B) Adds a CPU per app
C) Closes every other app
D) Gives each app all RAM

3. "Open source means there are never any licence conditions or costs." Correct this. Explain why.

4. A school uses biased software to choose pupils. A business wants to share licensed software. Correct both claims. Explain why. (a) Every legal use is ethical. (b) Law depends only on what is popular.

Today's objectives

Show how insertion sort builds a sorted portion. Trace values moving right to insert a key.

Key vocabulary - Read, cover, recall

Insertion sort
Builds a sorted section by inserting one value at a time.

Key
Value currently being inserted.

Sorted portion
Part of the array already in ascending or descending order.

Shift
Move a value one position to make room.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Sorted prefix	Insertion sort begins with the first item treated as a sorted portion.
Choose a key	Take the next unsorted value and compare it with values already in the sorted portion.
Shift larger entries	Move larger sorted values one position right to make a gap; do not lose the saved key.
Insert and repeat	Place the key into the gap and extend the sorted portion until the array is fully ordered.

I Do - One worked example

Trace insertion sort on [4,1,3] and give the array after each insertion. [4 marks]

1. Initially [4] is treated as sorted [1].

2. Insert key 1 before 4, shifting 4 right: [1,4,3] [1].

3. Next key 3 is compared with 4 and 4 shifts right [1].

4. Insert 3 after 1 to form [1,3,4] [1].

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Apply insertion sort to [5,2,4]. Show the array after inserting 2 and after inserting 4. [4 marks]

2. Explain what happens when inserting 7 into sorted prefix [2,5,8]. [3 marks]

3. Explain one difference between insertion and bubble sorts in how they reorder data. [4 marks]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Fill the missing swap assignment in the first insertion step.

```
array sizes = [4,1,3]
key = sizes[1]
sizes[1] = sizes[0]
... = key
print(sizes[0])
print(sizes[1])
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State what is meant by the key in insertion sort. [2 marks]

Q3. Show insertion sort on [6,3,5] after each insertion. [4 marks]

Q4. Explain how insertion sort orders [7,2,5,1] by maintaining a sorted section and shifting values. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Insertion sort inserts each new key into the correct position of a growing sorted part. ☐ T ☐ F

2. Insertion sort begins with the first item treated as a sorted portion. ☐ T ☐ F

3. Take the next unsorted value and compare it with values already in the sorted portion. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 When the left item is greater than the right item. ☐

Q2 A - Shares CPU time ☐

Q3 It permits access to source code and modification or redistribution under its conditions. ☐

Q4 (a) Ethics concerns fairness, responsibility and what is right. (b) Law sets what is allowed or banned. Consider effects on people: a legal action may still be unfair or intrusive. ☐

Code First - Model

```
array sizes = [4,1,3]
key = sizes[1]
sizes[1] = sizes[0]
sizes[0] = key
print(sizes[0])
print(sizes[1])
```

Output: 1
4

We Do - Check the model

1. Treat 5 as sorted [1]; shift 5 right and insert 2: [2,5,4] [1]. Move 5 right [1]; insert 4 to get [2,4,5] [1]. ☐

2. Compare 7 with 8 [1], shift 8 one position right [1], place 7 after 5 and before 8 [1]. ☐

3. Insertion sort takes a key [1] and inserts it into a growing sorted section [1]. Bubble sort compares adjacent pairs [1] and swaps out-of-order neighbours over passes [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 The current unsorted element being inserted [1] into its correct place among the already-sorted elements [1]. ☐

Q3 Initially 6 is sorted [1]. Shift 6 and insert 3 to produce [3,6,5] [1]. Shift 6 again [1] and insert 5 to get [3,5,6] [1]. ☐

Q4 Start with [7] as sorted [1]. Insert 2 before 7 -> [2,7,5,1] [1]. Insert 5 before 7 -> [2,5,7,1] [1]. For key 1 shift 7 then 5 then 2 right [2]. Insert 1 first -> [1,2,5,7] [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: